Shuying Cao

LLM Agents & Evaluation, Trustworthy AI, AI for Science & Scientific Discovery, RSI
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EDUCATION

University of Southern California (USC) — **Los Angeles, CA, US**
M.S. in Computer Science, GPA 3.78/4.0 — *Sept. 2024 – Dec. 2026 (expected)*

Stanford University, *Visiting Student, Summer Session* — *June 2026 – Aug. 2026*

University of California, Berkeley (UCB), *Visiting Student, SKY Lab* — *Jan. 2023 – Jul. 2023*

Wuhan University (WHU) — **Wuhan, Hubei, China**
B.E. in Geodesy and Geomatics Engineering, GPA 3.81/4.0 (rank **top 2%**) — *Sept. 2020 – June 2024*

- Qinghai Li Scholarship (top 2%, 2022); First Class Scholarship of Wuhan University (2021 & 2022).

PUBLICATIONS (* equal contribution)

[1] Jacob Choi*, **Shuying Cao***, Xingjian Dong*, Amin Banayeeanzade, Wang Bill Zhu, Robin Jia, Sai Praneeth Karimireddy. **ContextLeak: Auditing Leakage in Private In-Context Learning Methods**. COLM 2026. arXiv

[2] **Shuying Cao***, Yixin Wang*, et al., Yejin Choi, Michael Zeineh. NeuroImAgent: An AI Agent for Translating Pathology to Neuroradiology. In preparation.

[3] Lihao Sun, Shan Gao, **Shuying Cao**, et al. Towards Intent-Aware Language Models. In preparation.

RESEARCH EXPERIENCE

Agents' Last Exam (ALE) v2 – SWE & Science Track — *Oct. 2026 – Present*
Core Contributor, BAIR, UC Berkeley

- SWE Track: collect real software behavior as ground truth and turn it into examples for agents; own the data collection and the evaluation of whether agent-built software reproduces the original behavior and interface.
- Science Track: own evaluation for the self-improvement loop. Mapped which stages of the scientific process existing open benchmarks cover (459 benchmarks across 88 subfields, 2.4K–4.7K reusable agentic tasks) and am synthesizing tasks for the uncovered stages so an agent can run the full discovery loop end to end, graded at each step.

NeuroImAgent – Agent for Translating Pathology to Neuroradiology — *June 2026 – Present*
*RA, advised by **Prof. Michael Zeineh** and in collaboration with **Prof. Yejin Choi**, Stanford University*

- An AI agent system that analyzes brain MRI on its own: it takes raw scans, organizes them into BIDS format, runs neuroimaging pipelines, and returns the measurements researchers need to connect pathology with imaging.
- Designed the agent to improve itself from experience: every case is stored in a memory graph linking scan protocols, decisions, tool calls, and outcomes, so new subjects are handled by retrieving similar past cases; procedures that work are saved as reusable skills and refined when they fail.
- Made it safe and reproducible on patient data: the agent runs in a Bubblewrap + Docker sandbox, a harness keeps its behavior consistent and every result in one structured format (336 tests), and memory is versioned per run. Integrated large in vivo and ex vivo MRI datasets for biomarker studies.

ContextLeak: Black-box Privacy Auditing of Private ICL — *Dec. 2024 – May 2025*
*RA, advised by **Prof. Sai Praneeth Karimireddy**, FORT-ML Lab, USC*

- Built ContextLeak, the first black-box framework to measure worst-case leakage in private in-context learning: insert a canary into the exemplars with probability 1/2, craft a query whose output reveals it, and turn auditor accuracy into an empirical lower bound on ε (no logits or gradients needed).
- Designed a search for the strongest attack per defense: 3 canary types & 3 query strategies, refined with GEPA; LLM-based defenses block prompt injection but not ContextLeak.
- Showed that empirical ε tracks theoretical ε for DP defenses, that ensembling adds privacy even at $\varepsilon = \infty$, and that utility plateaus early with a persistent gap to non-private ICL.

Multimodal Chatbot for Interpreting Figures in Academic Papers — *May 2023 – Sept. 2023*
*RA, advised by **Prof. Joseph E. Gonzalez**, PhD Student Tianjun Zhang, SKY Lab, UC Berkeley*

- Built a LLaVA-style vision–language assistant on LLaMA 2 / Alpaca that answers questions about figures and plots in research papers, helping researchers read scientific literature.
- Generated and refined the figure–question–answer instruction-tuning data with GPT-4.

PROJECTS

Soul Soul – Persona Layer for OpenClaw Agents (soulsoul.live) — *Mar. 2026 – Present*

- Led the "soul" spec mapping persona attributes to structured prompts and behavior; 4K+ page views in 3 days.

SKILLS

- ML / LLM: PyTorch, Transformers, LLM Agents & Tool Use, RAG, In-Context Learning, LLM Evaluation;
- Data & Research: Experiment Design, Statistical Analysis, Benchmarking, Privacy Auditing;